

Vedant Naidu

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Two years ago an accident left me with optic neuropathy, and navigating recovery showed me how often healthcare systems act without a complete picture of the patient. Studying the Dayflow codebase taught me that production AI is a context-engineering problem: deciding what information to collect, how to structure it, and when to surface it. I am applying that lesson to a smart-glasses health companion that pairs first-person observations with wearable biometrics so AI can reason over both behavior and the body's response.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js/Express, FastAPI, React Native, Tailwind CSS, LangChain, LangGraph

Tools & Infra: AWS (CDK, Lambda, Bedrock, OpenSearch, DynamoDB, Amplify, S3), PostgreSQL, Supabase, Firebase, MongoDB, Docker, Git, GitHub Actions, Kafka, Cursor, Claude Code

AI/ML: Voice AI Agents, LLMs (OpenAI & Anthropic APIs), Agentic & Multi-Agent Workflows, RAG Pipelines, Context Engineering, LLM Evals, Amazon Bedrock, Computer Vision, Whisper, scikit-learn

EDUCATION

University of Massachusetts Amherst

Expected 2027

Bachelor of Science in Computer Science

Amherst, MA

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer System Principles, Computer Architecture, Computational Statistics, Human-Centered Design

EXPERIENCE

Artificial Intelligence Engineering Intern

Jan 2026 – Present

Commonwealth of Massachusetts (DPH/EOHHS)

Boston, MA

- Shipped a production RAG chatbot used by **351 Massachusetts health departments** that cut CMR regulatory lookup from **8+ minutes to under 30 seconds** and handles **500+ inspector queries per week**, returning cited, guardrailed answers grounded entirely in the source regulatory PDFs with zero hallucinated citations
- Implemented contextual retrieval by replacing a wide cross-section chunker that mislabeled **~85% of chunks** with section-bounded ~110-token chunks carrying a template context anchor, lifting per-citation recall from **0.71 to 0.86** while feeding the generator **~5× fewer tokens** per query across the full evaluation set
- Built hybrid semantic+keyword search with intent-based routing (definitional queries to semantic, formula queries to table-specific chunks) and a Claude Haiku cross-encoder rerank stage, validated at **94% retrieval precision** on a curated 100-question ground-truth LLM evaluation benchmark spanning all three regulatory inspection domains

Computational Biology Researcher

Sep 2025 – Dec 2025

Stanford University (Optic Nerve Regeneration Lab)

Stanford, CA / Remote

- Built a Python pipeline processing 3,000+ protein-interaction records from public databases, surfacing **12 high-confidence wet-lab candidates** and cutting the experimental search space by **99.6%** before any wet-lab spend
- Integrated graph-centrality signals with sequence-embedding similarity into a unified scoring framework eliminating **95% of low-signal candidates**; the full pipeline reruns candidate selection on any new database snapshot in **under 10 minutes**

PROJECTS

Elderly Health Monitoring Voice Agent

GitHub | *Best Healthcare Hack, HackRPI*

- Built an AI voice agent that uses patient medications and conditions to guide adaptive wellness check-ins, converting each conversation into structured Supabase records and automating caregiver outreach with **98% call-completion reliability**
- Designed tool-calling guardrails that capture medication adherence, mood, sleep, and symptoms as validated fields, requiring the agent to resolve missing information before generating the final patient summary
- Connected post-call webhooks to a Node.js and FastAPI anomaly pipeline that validates audio quality, compares voice embeddings against patient-specific baselines, and escalates warning or emergency signals within **10 seconds**

Voice Maintenance Agent

GitHub | *Best Use of AI, HackUMass*

- Built a multi-turn voice agent using Whisper, GPT-4, and speech synthesis to replace rigid maintenance forms with natural conversation, generating a **100+ person waitlist** from live demonstrations
- Used LLM function calling to produce schema-validated issue type, location, urgency, routing category, summary, and missing-field state, keeping the conversation active until the downstream ticket was complete
- Streamed validated outputs through an Express and Firebase backend that routed Maintenance versus Resident Life requests and synchronized urgency-and-time-ranked tickets across resident, RA, and staff dashboards

GymGlasses: Context-Aware Smart-Glasses Coach

GitHub

- Building a hands-free lifting coach that uses first-person video from Ray-Ban Meta glasses to track exercise progress and deliver spoken form guidance without requiring the athlete to hold or check a phone
- Engineered a rolling context layer that reduces the video stream to keyframes and combines them with exercise type, rep history, prior coaching, and spoken feedback before each multimodal model request
- Built a test-first Swift core for frame sampling, rep counting, session-state transitions, response parsing, and cue throttling, isolating model and hardware integrations behind protocols for independent testing